Qian 2020 Ovarian — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=20) interleaves all 4 patients: same-sample neighbour fraction 1.00 → 0.63 (chance 0.35), iLISI median 1.00 → 2.13 of 4. θ=30 REVERSES mixing to 0.88 (over-integration fragments into per-sample lobes), so θ=20 is the strongest integration on the plateau. 97 cells / 1 detached island removed → one connected component. Read on the regenerated embedding + iLISI: mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

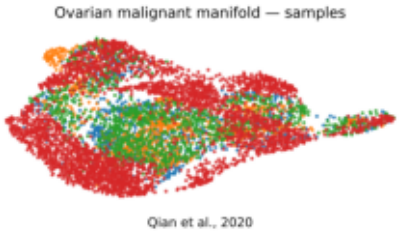
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-express (joint enrichment 1.20 > 1; smoothed r=+0.14), and both oppose Crest (Gast↔Crest r=−0.28, pSom↔Crest r=−0.26; joint enrichment 0.58 / 0.63 < 1 = inverse, not a co-expression hub). Along pseudotime Gast/pSom fall (ρ −0.46 / −0.55), Crest rises to a terminal tip (high-decile pt = 0.83).

Gate 3 — hub cross-repression → NA

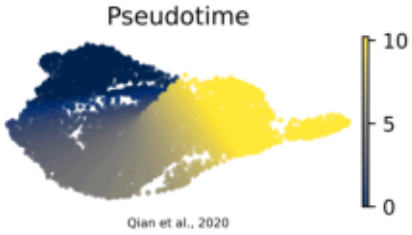
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks, fold self-approach 2.41). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

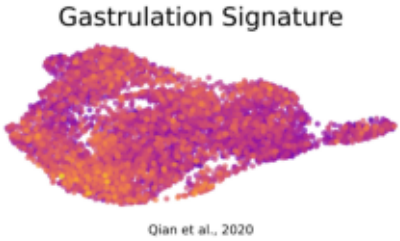
Manifold — samples



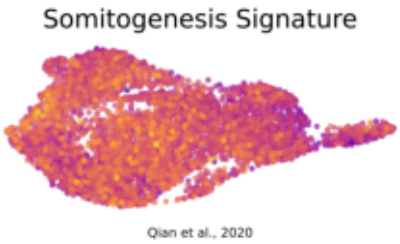
Pseudotime



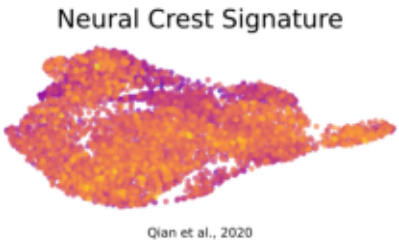
Gastrulation



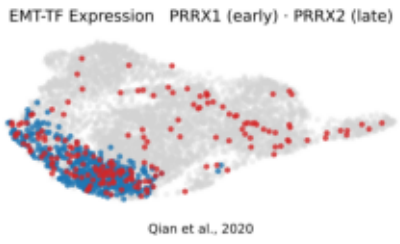
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Ovarian cancer (Qian et al., 2020), 10x — 6,393 malignant cells over 4 treatment-naive patients; one continuous low-confound body (embryonic modules proliferation-independent).

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs PRRX1 (early/root) → PRRX2 (late/terminus).

Parameter choices

Sample threshold	≥100 cells (floor; no dominant cliff, ratio 1.89) → 4/4 samples, 6,490 cells	**Overdispersed genes**	1,013 (scFates find_overdispersed)
Integration	Harmony on sample, θ=20 (mixing plateau; θ=30 over-integrates and fragments)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 97 cells / 1 island removed → 6,393
Geometry	Non-branching → scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.41	**Root**	tip 0 = Gast/pSom-high, Crest-low origin; corr(x_lin,pt)=+0.83
Module means (raw)	Gast +0.042 · pSom +0.065 · Crest −0.019 (Linear-pSom/Crest)	**EMT-TFs shown**	PRRX1 (early/root, ρ −0.19) ↔ PRRX2 (late/terminus, ρ +0.00); detection-gated